

# LINUX ARCHIVES, COMPRESSION AND FILE TRANSFER

Manage, compress, archive and securely move files in Linux to save space, organize data and work across systems efficiently.



## KEY CONCEPTS



### ARCHIVE

A single file that contains one or more files or directories.



### COMPRESSION

Reduces file size to save storage and bandwidth.



### FILE TRANSFER

Moves files between systems, locally or over a network.



### SECURITY

Use secure methods (scp, sftp, rsync over SSH) to protect data in transit.



Archiving, compression and secure file transfer are essential day-to-day DevOps skills.

## COMPRESSION COMMANDS

| COMMAND | DESCRIPTION                 | EXAMPLE              |
|---------|-----------------------------|----------------------|
| gzip    | Compress file (create .gz)  | gzip file.txt        |
| gunzip  | Decompress .gz              | gunzip file.txt.gz   |
| bzip2   | Compress file (create .bz2) | bzip2 file.txt       |
| bunzip2 | Decompress .bz2             | bunzip2 file.txt.bz2 |
| xz      | Compress file (create .xz)  | xz file.txt          |
| unxz    | Decompress .xz              | unxz file.txt.xz     |



For a single file, use gzip (fast) or xz (better compression).

## ARCHIVE COMMANDS

| COMMAND   | DESCRIPTION                         | EXAMPLE                           |
|-----------|-------------------------------------|-----------------------------------|
| tar -cvf  | Create archive                      | tar -cvf backup.tar /etc          |
| tar -xvf  | Extract archive                     | tar -xvf backup.tar               |
| tar -czvf | Create compressed archive (.tar.gz) | tar -czvf backup.tar.gz /var/log  |
| tar -xzvf | Extract .tar.gz                     | tar -xzvf backup.tar.gz           |
| tar -cjvf | Create .tar.bz2                     | tar -cjvf backup.tar.bz2 /var/www |
| tar -xjvf | Extract .tar.bz2                    | tar -xjvf backup.tar.bz2          |



Use tar with gzip or bzip2 to archive and compress directories in one step.

## SCP – Secure Copy

Securely copy files and directories over SSH.

| COMMAND                            | DESCRIPTION / EXAMPLE        |
|------------------------------------|------------------------------|
| scp file.txt user@server:/path/    | Copy file to remote server   |
| scp user@server:/path/file.txt .   | Copy file from remote server |
| scp -r project/ user@server:/path/ | Copy directory recursively   |



Local Machine



Remote Server

## RSYNC – Fast and Efficient

Synchronize files and directories with speed and resume support.

| COMMAND                              | DESCRIPTION / EXAMPLE         |
|--------------------------------------|-------------------------------|
| rsync -av source/ dest/              | Sync directories locally      |
| rsync -av source/ user@server:/path/ | Sync to remote server         |
| rsync -av user@server:/path/ dest/   | Sync from remote server       |
| rsync -av --delete source/ dest/     | Sync and delete removed files |



Fast • Incremental • Reliable



## SFTP – Secure File Transfer

Interactive and secure file transfer over SSH.

| COMMAND               | DESCRIPTION / EXAMPLE        |
|-----------------------|------------------------------|
| sftp user@server      | Connect to remote server     |
| sftp> put file.txt    | Upload file to server        |
| sftp> get file.txt    | Download file from server    |
| sftp> put -r project/ | Upload directory recursively |
| sftp> bye             | Exit sftp session            |



Encrypted Connection



## PRACTICAL EXAMPLES

1

### Create and extract a compressed backup

```
tar -czvf site-backup.tar.gz /var/www
tar -xzvf site-backup.tar.gz
```

2

### Copy a file to a remote server

```
scp site-backup.tar.gz ubuntu@192.168.1.10:/home/ubuntu/
```

3

### Sync project directory using rsync

```
rsync -av --delete ./app/ ubuntu@192.168.1.10:/var/www/app/
```

4

### Upload multiple files with sftp

```
sftp ubuntu@192.168.1.10
sftp> put -r ./reports
```

## ARCHIVE & COMPRESSION FORMAT COMPARISON

| FORMAT  | EXTENSION | BEST FOR                           | COMMAND    |
|---------|-----------|------------------------------------|------------|
| tar     | .tar      | Archiving (no compression)         | tar -cvf   |
| tar.gz  | .tar.gz   | General purpose (small size, fast) | tar -czvf  |
| tar.bz2 | .tar.bz2  | Better compression                 | tar -cjvf  |
| tar.xz  | .tar.xz   | Best compression ratio             | tar -cJf   |
| zip     | .zip      | Cross-platform sharing             | zip /unzip |
| gzip    | .gz       | Single files (fast)                | gzip       |
| bzip2   | .bz2      | Single files (better ratio)        | bzip2      |
| xz      | .xz       | Single files (best ratio)          | xz         |

## BEST PRACTICES

- ✓ Use tar with gzip/xz for backups.
- ✓ Prefer rsync for large directories or frequent syncs.
- ✓ Use scp or sftp for secure transfers over SSH.
- ✓ Verify files after transfer using checksum (md5sum or sha256sum).
- ✓ Automate backups and transfers with shell scripts or cron.
- ✓ Keep sensitive files encrypted in transit and at rest.



## QUICK TIP

Verify file integrity after transfer:

```
sha256sum file.txt
```

Compare the checksum on both source and destination.



# LINUX DISK AND STORAGE MANAGEMENT



Linux provides powerful tools to manage disks, partitions, filesystems and storage efficiently and safely.

## KEY CONCEPTS



### DISK

A physical storage device (HDD, SSD, NVMe, USB, etc.)



### PARTITION

A division of a disk. Each partition works like a separate disk.



### FILE SYSTEM

Organizes data on a partition (ext4, xfs, btrfs, etc.).



### MOUNT POINT

Directory where a file system is attached to the Linux directory tree.



### STORAGE MANAGEMENT

Includes partitioning, formatting, mounting, monitoring and maintaining disks.

## CHECK DISKS AND PARTITIONS

| COMMAND              | EXAMPLE              | DESCRIPTION                                              |
|----------------------|----------------------|----------------------------------------------------------|
| lsblk                | lsblk -f             | List block devices and partitions (with filesystem info) |
| blkid                | blkid                | Show UUIDs and filesystem types                          |
| fdisk                | sudo fdisk -l        | List partition table (MBR/GPT)                           |
| parted               | sudo parted -l       | List disks and partitions                                |
| df                   | df -h                | Show disk space usage of mounted filesystems             |
| du                   | du -sh /path         | Show disk usage of files/directories                     |
| cat /proc/partitions | cat /proc/partitions | Show kernel partition information                        |
| mount                | mount   column -t    | Show mounted filesystems                                 |

**i** lsblk shows the relationship between disks, partitions and mount points.

## PARTITIONING DISKS

| TASK                     | COMMAND (EXAMPLE)   | DESCRIPTION                            |
|--------------------------|---------------------|----------------------------------------|
| Open fdisk               | sudo fdisk /dev/sdb | Open disk for partitioning             |
| Create new partition (n) | n                   | Create new partition                   |
| Delete partition (d)     | d                   | Delete a partition                     |
| Write changes            | w                   | Save changes and exit                  |
| Quit without saving      | q                   | Discard changes and quit               |
| View partition table     | p                   | Print partition table                  |
| Change partition type    | t                   | Change partition type (MBR)            |
| With GPT, use gdisk      | sudo gdisk /dev/sdb | GPT partitioning tool (better for GPT) |

**i** Always backup important data before partitioning.  
Use GPT for disks > 2TB and modern systems.

## FILESYSTEM MANAGEMENT

| COMMAND                 | EXAMPLE                      | DESCRIPTION                 |
|-------------------------|------------------------------|-----------------------------|
| Make filesystem (ext4)  | sudo mkfs.ext4 /dev/sdb1     | Create ext4 filesystem      |
| Make filesystem (xfs)   | sudo mkfs.xfs /dev/sdb1      | Create XFS filesystem       |
| Make filesystem (btrfs) | sudo mkfs.btrfs /dev/sdb1    | Create Btrfs filesystem     |
| Check filesystem        | sudo fsck /dev/sdb1          | Check and repair filesystem |
| Check ext4 (specific)   | sudo e2fsck -f /dev/sdb1     | Force check ext4 filesystem |
| Resize ext4             | sudo resize2fs /dev/sdb1     | Resize ext4 filesystem      |
| Resize XFS              | sudo xfs_growfs /mount/point | Grow XFS filesystem         |
| Label filesystem        | sudo e2label /dev/sdb1 DATA  | Set filesystem label        |
| UUID of filesystem      | sudo blkid /dev/sdb1         | Show UUID of filesystem     |

**i** Choose the right filesystem: ext4 (general), xfs (performance), btrfs (advanced features).

## MOUNTING AND UNMOUNTING

| COMMAND              | EXAMPLE                                | DESCRIPTION                |
|----------------------|----------------------------------------|----------------------------|
| Create mount point   | sudo mkdir /mnt/data                   | Create directory for mount |
| Mount filesystem     | sudo mount /dev/sdb1 /mnt/data         | Mount device to directory  |
| Mount with options   | sudo mount -o ro /dev/sdb1 /mnt/data   | Mount read-only            |
| Unmount              | sudo umount /mnt/data                  | Unmount filesystem         |
| View mounts          | mount   column -t                      | Show mounted filesystems   |
| Mount all from fstab | sudo mount -a                          | Mount all listed in fstab  |
| Temporary mount      | sudo mount -t ext4 /dev/sdb1 /mnt/data | Mount one time only        |

**i** Edit /etc/fstab for permanent mounts.

```
# <device>      <mount point> <type>  <options>  <dump>  <pass>
UUID=1234-ABCD /mnt/data     ext4     defaults   0        2
```

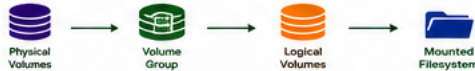
Options examples: defaults, ro, rw, noatime, nodev, nosuid

## DISK USAGE AND MONITORING

| COMMAND                        | EXAMPLE                          | DESCRIPTION                       |
|--------------------------------|----------------------------------|-----------------------------------|
| Disk usage summary             | df -h                            | Human readable disk usage         |
| Inodes usage                   | df -i                            | Show inode usage                  |
| Show disk usage by type        | df -Th                           | Show filesystem types             |
| Show largest files/directories | du -ah /   sort -hr   head -n 10 | Top 10 largest in /               |
| Watch disk usage               | watch -n 2 df -h                 | Refresh disk usage every 2 sec    |
| SMART info (HDD/SSD)           | sudo smartctl -a /dev/sda        | Check disk health (smartmontools) |
| Disk performance               | iostat -xz 1                     | I/O stats (sysstat package)       |
| Real-time I/O                  | iotop                            | Show top I/O using processes      |

**i** Install tools: sudo apt install smartmontools sysstat iotop

## LVM (LOGICAL VOLUME MANAGER)



| TASK             | COMMAND (EXAMPLE)                        | DESCRIPTION                    |
|------------------|------------------------------------------|--------------------------------|
| Create PV        | sudo pvcreate /dev/sdb1                  | Initialize physical volume     |
| Create VG        | sudo vgcreate vgdata /dev/sdb1           | Create volume group            |
| Create LV        | sudo lvcreate -L 20G -n lvdata vgdata    | Create logical volume (20G)    |
| View LVs         | lvdisplay                                | Display logical volume info    |
| Format LV        | sudo mkfs.ext4 /dev/vgdata/lvdata        | Create filesystem on LV        |
| Mount LV         | sudo mount /dev/vgdata/lvdata /mnt/data  | Mount logical volume           |
| Extend LV        | sudo lvextend -L +10G /dev/vgdata/lvdata | Extend logical volume          |
| Extend FS (ext4) | sudo resize2fs /dev/vgdata/lvdata        | Resize filesystem after extend |

**i** LVM allows flexible resizing and management of storage.

## RAID OVERVIEW (mdadm)

| LEVEL                      | DESCRIPTION                                                         | USE CASE                              |
|----------------------------|---------------------------------------------------------------------|---------------------------------------|
| RAID 0 (Striping)          | Data striped across disks. No redundancy.                           | High performance                      |
| RAID 1 (Mirroring)         | Data mirrored on 2 disks. 1 disk can fail.                          | Redundancy                            |
| RAID 5 (Striping + Parity) | Data striped with parity. 1 disk can fail.                          | Balance of performance and redundancy |
| RAID 10 (1+0)              | Mirror + Stripe. High performance, 1 disk can fail per mirror pair. | High performance and redundancy       |

Basic mdadm commands

```
sudo mdadm --create /dev/md0 --level=1 --raid-devices=2 /dev/sdb1 /dev/sdc1
sudo mdadm --detail /dev/md0 # show raid details
sudo mdadm --stop /dev/md0 # stop array
```

**i** RAID protects against disk failure, not backups!

## BEST PRACTICES

- ✓ Always backup important data before making disk changes.
- ✓ Use SSD/NVMe for better performance.
- ✓ Use GPT partition table for modern systems.
- ✓ Use meaningful labels and UUIDs in /etc/fstab.
- ✓ Monitor disk health with smartctl regularly.
- ✓ Keep at least 10-20% free space on filesystems.
- ✓ Use LVM for flexibility in production environments.
- ✓ Test restores from backups periodically.



### QUICK TIP

Find UUID of a device:

```
sudo blkid /dev/sdb1
```

Use UUID in /etc/fstab instead of device name for stability.



### REMEMBER

Disk names like /dev/sdb can change. UUIDs don't!



# LINUX LOGS AND TROUBLESHOOTING

Logs tell you what happened.  
Troubleshooting gets you to  
the root cause and resolution.



## KEY CONCEPTS



### LOG

A record of events, errors, warnings and activities.



### CENTRALIZED LOGGING

Collect logs from multiple systems for easier analysis.



### TROUBLESHOOTING

Identify, isolate and resolve issues systematically.



### ROOT CAUSE

The underlying cause of a problem, not just the symptom.



### TOOLS & COMMANDS

Use the right tools to inspect, filter and analyze logs.

## COMMON LOG FILES AND LOCATIONS

| LOG FILE / PATH          | DESCRIPTION           | WHAT YOU WILL FIND                                            |
|--------------------------|-----------------------|---------------------------------------------------------------|
| /var/log/syslog          | General system logs   | Boot messages, system services, kern messages (Debian/Ubuntu) |
| /var/log/messages        | General system logs   | Boot messages, system services, kern messages (RHEL/CentOS)   |
| /var/log/secure          | Security & auth logs  | SSH logins, sudo, authentication events, PAM                  |
| /var/log/auth.log        | Authentication logs   | Logins, sudo, su, authentication (Debian/Ubuntu)              |
| /var/log/dmesg           | Kernel ring buffer    | Hardware, kernel, driver messages                             |
| /var/log/kern.log        | Kernel logs           | Kernel messages (older systems)                               |
| /var/log/boot.log        | Boot process logs     | System startup messages                                       |
| /var/log/daemon.log      | System daemons        | Background services and daemons                               |
| /var/log/maillog         | Mail server logs      | Postfix, Dovecot, sendmail logs                               |
| /var/log/faillog         | Failed login attempts | Failed logins with timestamps                                 |
| /var/log/cron            | Cron job logs         | Cron job execution details                                    |
| /var/log/yum.log         | Package manager logs  | Yum/DNF install, update, remove                               |
| /var/log/nginx/          | Nginx access/error    | Web server access & error logs                                |
| /var/log/audit/audit.log | Audit logs            | System audit events (if enabled)                              |

Log locations may vary by distribution and service configuration.

## VIEWING AND FILTERING LOGS

| COMMAND       | EXAMPLE                             | DESCRIPTION                                |
|---------------|-------------------------------------|--------------------------------------------|
| cat           | cat /var/log/syslog                 | Display entire log file                    |
| less          | less /var/log/syslog                | View log file page by page                 |
| tail          | tail /var/log/syslog                | Show last 10 lines                         |
| tail -f       | tail -f /var/log/syslog             | Follow log in real time                    |
| head          | head -n 20 /var/log/syslog          | Show first 20 lines                        |
| grep          | grep "error" /var/log/syslog        | Search for pattern                         |
| grep -i       | grep -i "fail" /var/log/secure      | Case-insensitive search                    |
| grep -v       | grep -v "debug" app.log             | Exclude pattern                            |
| awk           | awk '{print \$1, \$2, \$NF}' syslog | Print selected fields                      |
| sed           | sed -n '1,20p' syslog               | Print lines 1 to 20                        |
| journalctl    | journalctl -xe                      | View systemd logs (detailed, with context) |
| journalctl -u | journalctl -u nginx -f              | Follow logs for a service                  |

### Real-time monitoring example

```
tail -f /var/log/nginx/error.log | grep --line-buffered "error"
```

## TROUBLESHOOTING PROCESS



### 1 IDENTIFY THE PROBLEM

What is not working? Gather symptoms and user reports.



### 2 REPRODUCE THE ISSUE

Reproduce the problem if possible.



### 3 CHECK THE LOGS

Look for errors or warnings around the time of the issue.



### 4 ISOLATE THE CAUSE

Narrow down to the component causing the issue.



### 5 FIX THE PROBLEM

Apply the fix or change the configuration.



### 6 VERIFY THE FIX

Confirm the issue is resolved.



### 7 PREVENT RECURRENCE

Document and implement monitoring or alerts.

## SYSTEMD JOURNAL (JOURNALCTL)

| COMMAND                                                    | DESCRIPTION                             |
|------------------------------------------------------------|-----------------------------------------|
| journalctl                                                 | Show all logs                           |
| journalctl -f                                              | Follow logs in real time                |
| journalctl -xe                                             | Show logs with errors and context       |
| journalctl -u sshd                                         | Show logs for sshd service              |
| journalctl -u nginx -f                                     | Follow logs for nginx service           |
| journalctl --since "1 hour ago"                            | Show logs from the last 1 hour          |
| journalctl --since "2024-05-01" --until "2024-05-01 12:00" | Show logs between time range            |
| journalctl -p err                                          | Show logs with priority error and above |
| journalctl --disk-usage                                    | Show journal disk usage                 |
| journalctl --vacuum-time=7d                                | Keep logs for last 7 days               |

journalctl is the default log system on most modern Linux distributions using systemd.

## COMMON LOG PATTERNS

| PATTERN         | MEANING                                                 |
|-----------------|---------------------------------------------------------|
| ERROR           | Something went wrong. Needs attention.                  |
| WARN            | Potential issue. Investigate.                           |
| FAILED          | Operation or service failed.                            |
| TIMEOUT         | Operation took too long. Check network or performance.  |
| DENIED          | Access was denied. Check permissions or authentication. |
| STARTED/STOPPED | Service or process started/stopped.                     |

## COMMON TROUBLESHOOTING COMMANDS

| COMMAND           | EXAMPLE                 | DESCRIPTION                             |
|-------------------|-------------------------|-----------------------------------------|
| top / htop        | top                     | Check CPU & memory usage                |
| df -h             | df -h                   | Check disk space usage                  |
| free -h           | free -h                 | Check memory usage                      |
| uptime            | uptime                  | Check load average & uptime             |
| ip a / ip r       | ip a<br>ip r            | Show IP addresses<br>Show routing table |
| ss -tulnp         | ss -tulnp               | Check open ports & services             |
| ps aux            | ps aux   grep nginx     | Check running processes                 |
| systemctl status  | systemctl status nginx  | Check service status                    |
| systemctl restart | systemctl restart nginx | Restart a service                       |
| dig / nslookup    | dig google.com          | Check DNS resolution                    |
| ping              | ping 8.8.8.8            | Check network connectivity              |
| traceroute        | traceroute google.com   | Trace network path                      |

## LOG ROTATION

Logrotate manages log files to prevent disk space issues.

**CONFIG FILE**  
/etc/logrotate.conf

**CONFIG DIRECTORY**  
/etc/logrotate.d/

### COMMON DIRECTIVES

|            |                          |
|------------|--------------------------|
| daily      | Rotate daily             |
| weekly     | Rotate weekly            |
| compress   | Compress rotated logs    |
| rotate 7   | Keep last 7 rotated logs |
| missingok  | Ignore missing files     |
| notifempty | Do not rotate empty logs |

Check status: `logrotate -d /etc/logrotate.conf`  
Force rotate: `logrotate -f /etc/logrotate.conf`

## EXAMPLE: FIND AND ANALYZE AN ERROR

1 Find errors in the last 1 hour

```
journalctl -p err --since "1 hour ago"
```

2 Follow logs of a service

```
journalctl -u nginx -f
```

5 Check open ports

```
df -h
```

3 Search for failed SSH logins

```
grep "Failed password" /var/log/secure
```

4 Check disk space

```
ss -tulnp
```

## BEST PRACTICES

- Know where logs are stored on your system.
- Monitor logs in real time for critical systems.
- Set up log rotation to save disk space.
- Centralize logs for servers in production.
- Use meaningful alerts for errors and warnings.
- Regularly review logs for security and performance.



### QUICK TIP

Combine tools for powerful analysis:

```
journalctl -u nginx -f | grep --line-buffered "error"
```



### REMEMBER

Logs are your best friend when something breaks.  
Check logs first!





# LINUX CRON JOBS AND TASK SCHEDULING

Automate repetitive tasks,  
run commands at specific  
times or intervals, and  
improve system efficiency.



## WHAT IS CRON?

Cron is a time-based job scheduler in Linux/Unix. It runs commands or scripts automatically at specified times, dates, or intervals.

- Automate backups, reports, updates, log rotation, cleanups, etc.
- Runs in the background via the cron daemon.
- Crontab (cron table) defines scheduled tasks for a user.
- System-wide tasks are stored in /etc/crontab or /etc/cron.d/

## CRON SYNTAX

A cron schedule has 5 time fields followed by the command.

| *                | *              | *                         | *                                           | *                        | command_to_execute |
|------------------|----------------|---------------------------|---------------------------------------------|--------------------------|--------------------|
| Minute<br>(0-59) | Hour<br>(0-23) | Day of<br>Month<br>(0-12) | MoY of<br>[wso]<br>(0-17) (0 or 7 = Sunday) | Day of<br>Month<br>(0-7) |                    |

Example:

```
30 2 * * 1 /home/user/script.sh
At 02:30 AM every Monday
```

### SPECIAL CHARACTERS

- \* Any value
- , Value list separator (e.g., 1,3,5)
- Range of values (e.g., 1-5)
- / Step values (e.g., \*/15)
- @ Special strings (see table on the right)

### SPECIAL STRINGS (@)

|           |                                   |
|-----------|-----------------------------------|
| @reboot   | Run once at system startup        |
| @yearly   | Run once a year (0 0 1 1 *)       |
| @annually | Run once a year (same as @yearly) |
| @monthly  | Run once a month (0 0 1 * *)      |
| @weekly   | Run once a week (0 0 * * 0)       |
| @daily    | Run once a day (0 0 * * *)        |
| @midnight | Run once at midnight (0 0 * * *)  |
| @hourly   | Run once an hour (0 * * * *)      |

## EXAMPLES

| SCHEDULE     | MEANING                             |
|--------------|-------------------------------------|
| * * * * *    | Every minute                        |
| 0 * * * *    | Every hour at minute 0              |
| 0 0 * * *    | Every day at midnight               |
| 30 2 * * *   | Every day at 02:30 AM               |
| 0 9 * * 1-5  | Every weekday at 09:00 AM           |
| 0 0 * * 0    | Every Sunday at midnight            |
| 15 */2 * * * | Every 2 hours at minute 15          |
| */10 * * * * | Every 10 minutes                    |
| 0 0 1 * *    | First day of every month            |
| 0 0 */7 * *  | Every 7 days at midnight            |
| 45 23 15 * * | 11:45 PM on the 15th of every month |

## MANAGING CRON JOBS (CRONTAB)

| COMMAND              | DESCRIPTION                                   |
|----------------------|-----------------------------------------------|
| crontab -e           | Edit your user's crontab file                 |
| crontab -l           | List your current cron jobs                   |
| crontab -r           | Remove your crontab (all jobs)                |
| crontab -u <user> -e | Edit another user's crontab (requires sudo)   |
| crontab -u <user> -l | List another user's cron jobs (requires sudo) |
| crontab -i -r        | Remove crontab with confirmation              |

### Where are cron jobs stored?

- User crontab: /var/spool/cron/crontabs/<username>
- System crontab: /etc/crontab
- Additional jobs: /etc/cron.d/
- Hourly jobs: /etc/cron.hourly/
- Daily jobs: /etc/cron.daily/
- Weekly jobs: /etc/cron.weekly/
- Monthly jobs: /etc/cron.monthly/

## EXAMPLE: EDITING CRONTAB

1 Open crontab editor

```
crontab -e
```

2 Add a job (examples)

```
# Backup database every day at 01:30 AM
30 1 * * * /usr/bin/mysqldump -u root -p'mypass' db > /backup/db_$(date +%V).sql

# Clean log files every Sunday at 03:00 AM
0 3 * * 0 /usr/local/bin/clean_logs.sh

# Send report every weekday at 08:30 AM
30 8 * * 1-5 /usr/local/bin/send_report.sh
```

3 Save and exit

- In nano: Ctrl + O, Enter, Ctrl + X
- In vim: Esc, :wq, Enter

## VERIFICATION & TESTING

| COMMAND                           | DESCRIPTION                           |
|-----------------------------------|---------------------------------------|
| crontab -l                        | List scheduled jobs                   |
| systemctl status cron             | Check cron service status             |
| systemctl restart cron            | Restart cron service                  |
| journalctl -u cron                | View cron service logs (on systemd)   |
| tail -f /var/log/cron             | Monitor cron logs (on some distros)   |
| run-parts --test /etc/cron.daily/ | Test daily scripts without running    |
| /usr/bin/crontab -T               | Validate crontab syntax (RHEL/CentOS) |
| crontab -e                        | (check syntax shown on save)          |



Tip: Always test scripts manually before scheduling them.  
Use absolute paths in cron jobs.

## CRON LOGS

### Common log locations:

- /var/log/cron (Debian/Ubuntu)
- /var/log/cron.log (RHEL/CentOS)
- journalctl -u cron (systemd based)

### View recent cron logs

```
grep CRON /var/log/syslog
```

or

```
journalctl -u cron --since "1 day ago"
```

⚠ Ensure your scripts/logs handle output and errors. Redirect output if needed.

```
* * * * /path/script.sh >> /var/log/script.log 2>&1
```

## SYSTEM-WIDE CRON (EXAMPLES)

### /etc/crontab (system-wide jobs format)

```
# minute hour day month day_of_week user command
0 2 * * * root /usr/local/bin/system_task.sh
```

### /etc/cron.d/example (format)

```
# minute hour day month day_of_week user command
15 3 * * 1-5 www-data /usr/local/bin/website_backup.sh
```

- 1 Files in /etc/cron.d/ must have a username field. Make sure they have proper permissions (644).

## BEST PRACTICES

- ✓ Use absolute paths for all commands and scripts.
- ✓ Redirect output and errors to log files.
- ✓ Keep scripts idempotent and error-free.
- ✓ Test scripts manually before scheduling.
- ✓ Comment your crontab entries for clarity.
- ✓ Monitor cron logs regularly.
- ✓ Avoid running heavy tasks too frequently.

## COMMON USE CASES

- Automated backups
- Log rotation and cleanup
- Generate and email reports
- System health checks and alerts
- Database maintenance
- Update checks and deployments

## TROUBLESHOOTING

| ISSUE              | SOLUTION                                          |
|--------------------|---------------------------------------------------|
| Job not running    | Check cron service, logs, and syntax              |
| Permission denied  | Ensure script is executable and paths are correct |
| No output in logs  | Ensure redirection is set (>> logfile 2>&1)       |
| Environment issues | Set required variables in script or crontab       |
| Time mismatch      | Check system time and timezone                    |

## QUICK REFERENCE

|                     |                           |
|---------------------|---------------------------|
| Edit crontab        | crontab -e                |
| List jobs           | crontab -l                |
| Remove jobs         | crontab -r                |
| Cron service status | systemctl status cron     |
| Cron logs (Debian)  | tail -f /var/log/cron     |
| Cron logs (RHEL)    | tail -f /var/log/cron.log |

